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SECOND QUARTER EXAMINATION
Mathematics 7

Name: _____
Section: _____

Score: _____
Date: _____

Multiple Choice

Instructions: Read each item carefully. Encircle the letter of the best answer for each question.

- Which number is both a perfect square and a perfect cube?
A. 64
B. 72
C. 81
D. 125
- For a positive whole number m , when is $\sqrt{m}$ irrational?
A. When m is even
B. When m is greater than 10
C. When m is not a perfect square
D. When m is divisible by 3
- Which expression represents the cube root of a perfect cube and has a negative value?
A. $\sqrt{-64}$
B. $\sqrt[3]{-64}$
C. $\sqrt[3]{64}$
D. $\sqrt{64}$
- Which statement best explains why 12 is the square root of 144?
A. $12 + 12 = 24$
B. $144 - 12 = 132$
C. $12 \times 12 = 144$
D. $144 \div 2 = 72$
- Which number should be classified as irrational?
A. $\sqrt{81}$
B. $\sqrt[3]{-125}$
C. $\sqrt{72}$
D. $9/4$
- $\sqrt[3]{-75}$ is located between two consecutive integers?
A. 2 and 3
B. 3 and 4
C. 4 and 5
D. 5 and 6
- A square floor has an area of 225 m^2 . What is the perimeter of the floor?
A. 30 m
B. 45 m
C. 60 m
D. 90 m
- A cube-shaped box has a volume of 729 dm^3 . What is the length of one edge of the box?
A. 7 dm
B. 8 dm
C. 9 dm
D. 10 dm
- Where should $\sqrt{90}$ be plotted on the number line?
A. Between 8 and 9, closer to 9
B. Between 9 and 10, closer to 9
C. Between 9 and 10, closer to 10
D. Between 10 and 11, closer to 10
- A learner says, " $\sqrt{48}$ is rational because 48 is divisible by 16, a perfect square." Which explains the error?
A. $\sqrt{48}$ is rational because all even radicands are rational.
B. $\sqrt{48} = 4\sqrt{3}$ and since $\sqrt{3}$ is irrational, $\sqrt{48}$ irrational.
C. $\sqrt{48}$ is between 6 and 7, so it must be rational.
D. $\sqrt{48}$ cannot be simplified.
- Which arrangement is from least to greatest?
A. $\sqrt{5}, \sqrt[3]{8}, 2, 3, \sqrt{7}$
B. $\sqrt[3]{8}, \sqrt{5}, 2, 3, \sqrt{7}$
C. $\sqrt[3]{8}, 2, 3, \sqrt{7}, \sqrt{5}$
D. $2, 3, \sqrt[3]{8}, \sqrt{7}, \sqrt{5}$
- Which conversion factor should be used to convert inches to centimeters?
A. $\frac{2.54 \text{ cm}}{1 \text{ in}}$
B. $\frac{1 \text{ cm}}{2.54 \text{ in}}$
C. $\frac{12 \text{ in}}{1 \text{ ft}}$
D. $\frac{1 \text{ m}}{100 \text{ cm}}$



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13. Which situation best shows why standard units are preferred over non-standard units?
 - A. Measuring a desk using hand spans may give different answers for different learners.
 - B. Measuring a notebook using centimeters gives the same unit for everyone.
 - C. Using meters makes long distances easier to record.
 - D. Using liters is appropriate for measuring liquid volume.
14. When converting from a smaller metric unit to a larger metric unit, what should be done?
 - A. Multiply by a power of 10
 - B. Divide by a power of 10
 - C. Always multiply by 100
 - D. Always divide by 1,000
15. Which set contains only units used for capacity or volume?
 - A. meter, kilometer, centimeter, inch
 - B. gram, kilogram, pound, ounce
 - C. liter, milliliter, gallon, cup
 - D. square meter, cubic meter, foot, yard
16. Which pair contains one English unit and one metric unit used for mass or weight?
 - A. inch and meter
 - B. pound and kilogram
 - C. gallon and liter
 - D. foot and centimeter
17. Convert 0.075 km to centimeters.
 - A. 75 cm
 - B. 750 cm
 - C. 7,500 cm
 - D. 75,000 cm
18. Convert 3.75 kg to milligrams.
 - A. 3,750 mg
 - B. 37,500 mg
 - C. 375,000 mg
 - D. 3,750,000 mg
19. A ribbon is 6 feet long. How many meters long is the ribbon? Use $1 \text{ ft} = 0.3048 \text{ m}$.
 - A. 1.524 m
 - B. 1.8288 m
 - C. 2.4384 m
 - D. 3.048 m
20. A learner converted 4.6 kg to grams and wrote 460 g. Which statement best explains the mistake?
 - A. The learner multiplied by 100 instead of 1,000.
 - B. The learner divided by 100 instead of multiplying by 1,000.
 - C. The learner should have written 46 g.
 - D. The learner converted correctly.
21. A recipe needs 600 g of flour. Flour is sold in bags labeled in pounds. If $1 \text{ lb} \approx 454 \text{ g}$, what is the smallest bag that is enough?
 - A. 1 lb
 - B. 1.25 lb
 - C. 1.5 lb
 - D. 2.5 lb
22. Which formula correctly gives the volume of a cylinder when the diameter is d and the height is h ?
 - A. $V = \pi dh$
 - B. $V = \pi d^2 h$
 - C. $V = \pi \left(\frac{d}{2}\right)^2 h$
 - D. $V = 2\pi dh$
23. In finding the volume of a pyramid, which height must be used?
 - A. The slant height along a triangular face
 - B. The perpendicular height from the vertex to the base
 - C. The length of one base edge
 - D. The distance around the base
24. Why is the factor $\frac{1}{3}$ used in the volume formula of a pyramid?
 - A. A pyramid always has three sides.
 - B. The base of every pyramid is a triangle.
 - C. Three pyramids with the same base and height can fill a corresponding prism.
 - D. The height of a pyramid is always one-third of its base length.
25. Which formula is used to find the volume of a square pyramid with base edge s and height h ?
 - A. $V = s^2 h$
 - B. $V = \frac{1}{2} s^2 h$
 - C. $V = \frac{1}{3} s^2 h$
 - D. $V = 3s^2 h$
26. A right circular cylinder has _____.



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- A. two congruent and parallel circular bases
B. one circular base and one vertex
C. four triangular faces
D. a rectangular base and circular sides
27. Which information is not enough by itself to determine the volume of a cylinder?
A. radius and height
B. diameter and height
C. circumference of the base and height
D. height only
28. Find the volume of a cylinder with diameter 8 cm and height 10 cm.
A. 251.2 cm^3
B. 401.92 cm^3
C. 502.4 cm^3
D. $2,009.6 \text{ cm}^3$
29. Find the volume of a rectangular pyramid with length 15 cm, width 8 cm, and height 12 cm.
A. 360 cm^3
B. 420 cm^3
C. 480 cm^3
D. $1,440 \text{ cm}^3$
30. A square pyramid has a base edge of 14 cm and a volume of 588 cm^3 . What is its height?
A. 6 cm
B. 8 cm
C. 9 cm
D. 12 cm
31. Container A is a cylinder with radius 3 cm and height 12 cm. Container B is a square pyramid with base edge 12 cm and height 9 cm. Which statement is correct?
A. Container A has the greater volume by 92.88 cm^3 .
B. Container B has the greater volume by 92.88 cm^3 .
C. The containers have the same volume.
D. Container B has the greater volume by 432 cm^3 .
32. A student finds the volume of a rectangular pyramid with length 6 cm, width 5 cm, perpendicular height 8 cm, and slant height 10 cm. The student uses 10 cm as the height and gets 100 cm^3 . What error did the student make?
A. The student should have multiplied by 3.
B. The student used the slant height instead of the perpendicular height.
C. The student forgot to multiply the length and width.
D. The student used the formula for a cylinder.
33. Which is a well-defined set?
A. The set of beautiful flowers
B. The set of tall learners in Grade 7
C. The set of multiples of 3 less than 20
D. The set of difficult subjects
34. Which statement about the empty set is true?
A. It has exactly one element.
B. It is a subset of every set.
C. It is always equal to the universal set.
D. It contains zero as its only element.
35. What is the cardinality of the set of letters in the word "MATHEMATICS"?
A. 7
B. 8
C. 9
D. 11
36. A proper subset of a set is _____.
A. a subset that is equal to the original set
B. a subset that has at least one element not in the original set
C. a subset that is not equal to the original set
D. a subset that contains all possible elements in the universal set
37. In a Venn diagram, the overlapping region of two circles represents the _____.
A. union of two sets
B. intersection of two sets
C. universal set
D. empty set only
38. Let $A = \{\text{factors of } 24\}$ and $B = \{\text{even numbers less than } 15\}$. What is $A \cap B$?
A. $\{1, 2, 3, 4, 6, 8, 12, 24\}$
B. $\{2, 4, 6, 8, 10, 12, 14\}$
C. $\{2, 4, 6, 8, 12\}$
D. $\{10, 14, 24\}$
39. In a class of 30 learners, 18 joined the Math Club, 16 joined the English Club, and 5 joined neither club. How many learners joined both clubs?
A. 7
B. 8
C. 9
D. 10



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40. Let $A = \{1, 3, 5, 7, 9\}$ and $B = \{\text{prime numbers less than } 10\}$. What is the cardinality of $A \cup B$?
- A. 4
B. 5
C. 6
D. 7
41. Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5\}$, and $C = \{1, 2\}$. Which statement correctly explains the relationship among the sets?
- A. $C \subseteq B$ because C has two elements.
B. $C \subseteq A$ because every element of C is in A, but $C \not\subseteq B$ because 1 and 2 are not in B.
C. $A \subseteq C$ because A has more elements than C.
D. $B \subseteq A$ because 3 and 4 are common to both sets.
42. In a survey, 20 learners like basketball, 17 like volleyball, and 8 like both. How many learners like exactly one of the two sports?
- A. 21
B. 25
C. 29
D. 37
43. Among the sets N , W , Z , Q and R , what is the smallest set that contains -7 ?
- A. W
B. Z
C. Q
D. R
44. In the lesson on real numbers, which symbol represents irrational numbers?
- A. N
B. W
C. Q
D. Q'
45. Which statement correctly describes the relationship among subsets of real numbers?
- A. $W \subset Z \subset Q \subset R$
B. $Q' \subset Q$
C. $Z \subset W \subset N$
D. $R \subset Q$
46. Which subsets of real numbers contain 0?
- A. N , W , Z , Q , R
B. W , Z , Q , R
C. Z , Q' , R
D. Q' , R
47. Which number belongs to Q' and R ?
- A. $\sqrt{64}$
B. $\sqrt[3]{-27}$
C. $\sqrt[3]{9}$
D. $-\frac{5}{2}$
48. Arrange the numbers $-\frac{5}{2}$, $-\sqrt{6}$, -2.4 , 0 , and $\sqrt{2}$ in increasing order.
- A. $-\sqrt{6}$, $-\frac{5}{2}$, -2.4 , 0 , $\sqrt{2}$
B. $-\frac{5}{2}$, $-\sqrt{6}$, -2.4 , 0 , $\sqrt{2}$
C. -2.4 , $-\sqrt{6}$, $-\frac{5}{2}$, 0 , $\sqrt{2}$
D. $-\frac{5}{2}$, -2.4 , $-\sqrt{6}$, 0 , $\sqrt{2}$
49. Let Z be the set of integers and Q' be the set of irrational numbers. Which statement is true?
- A. $Z \cap Q' = \emptyset$
B. $Z \cup Q' = Z$
C. $Z \subset Q'$
D. $Q' \subset Z$
50. Which explanation proves that the statement "All real numbers are rational numbers" is false?
- A. Whole numbers are real numbers.
B. Integers are rational numbers.
C. Irrational numbers such as $\sqrt{2}$ are real numbers but not rational numbers.
D. Natural numbers are included in whole numbers.

"The only way to learn mathematics is to do mathematics." — Paul Halmos

Goodluck!



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ANSWER KEY:

- | | |
|-------|-------|
| 1. A | 26. A |
| 2. C | 27. D |
| 3. B | 28. C |
| 4. C | 29. C |
| 5. C | 30. C |
| 6. C | 31. B |
| 7. C | 32. B |
| 8. C | 33. C |
| 9. B | 34. B |
| 10. B | 35. B |
| 11. B | 36. C |
| 12. A | 37. B |
| 13. A | 38. C |
| 14. B | 39. C |
| 15. C | 40. C |
| 16. B | 41. B |
| 17. C | 42. A |
| 18. D | 43. B |
| 19. B | 44. D |
| 20. A | 45. A |
| 21. C | 46. B |
| 22. C | 47. C |
| 23. B | 48. B |
| 24. C | 49. A |
| 25. C | 50. C |

References:

DepEd Teachers Club. (n.d.). Welcome to DepEd Teachers Club! from <https://depedclub.com/>

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BSEd Mathematics 3